

Kunal Maheshwari

Mount Pleasant, MI, USA | (989) 906-2108 | kunal.maheshwari.work@gmail.com | <https://www.linkedin.com/in/kunalmaheshwari26>

SUMMARY

OPT-EAD holder with full-time work authorization and over 4 years of experience as a Software Developer and Data Engineer, seeking a Software Engineer II, Data Platform role. Possesses a Master of Science in Computer Science with expertise in building scalable microservices and distributed data pipelines, leveraging Java, Python, Spark, SQL, AWS, Azure, and GCP. Proven ability in CI/CD automation, infrastructure automation with Terraform, and developing privacy-preserving AI applications, with a strong understanding of HL7 and clinical care data models.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST APIs using Java 8/11, Spring Boot, and Azure Functions, handling over 50,000 daily API requests with 99.99 percent SLA uptime, integrating OIDC for secure authentication.
- Built and automated robust CI/CD pipelines using Jenkins, Maven, and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x and enhancing monitoring and error handling, reducing issue resolution time by 20 percent and improving data models.
- Led a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles for critical healthcare solutions.
- Collaborated within Agile and Scrum teams to deliver HL7-compliant healthcare solutions using Git, SVN, and JUnit testing, ensuring robust data integration.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered Java and EJB-based enterprise applications and RESTful web services for automotive clients, optimizing integration patterns.
- Improved system performance by 90 percent through SQL query optimization, database tuning, and modernization of legacy Java code, adhering to ANSI standards.
- Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates, supporting rapid deployment of cloud services.
- Mentored junior developers through code reviews and debugging sessions, promoting clean coding practices and a strong understanding of data models.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity through efficient ETL processes.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage and demonstrating expertise in Test Driven Development.
- Supported release planning and sprint execution using JIRA within Agile development teams, facilitating effective orchestration of tasks.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, GCP) January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze more than 8 GB of NYC taxi trip data in near real time, enhancing data integration capabilities.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization, leveraging REST APIs for data retrieval.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments across cloud services.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent and demonstrating efficient data processing.
- Automated cloud infrastructure provisioning using Terraform across AWS (S3, RDS, ECS, Lambda) and GCP services, enabling scalable data platforms.

Privacy-Preserving AI Email Client (Capstone Project, Python, Hugging Face) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, incorporating GenAI principles.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, demonstrating advanced AI techniques.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, ensuring privacy in clinical care contexts.
- Explored Linux environments for efficient model deployment and utilized Boost libraries for performance enhancements in data

processing.

SKILLS

- Programming Languages: Java, Python, SQL, Javascript.
- Frameworks and Libraries: Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit, React, Boost.
- Web Services: REST APIs, SOAP APIs, HL7.
- Cloud Platforms: AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform (GCP).
- Databases: PostgreSQL, Oracle Database, IBM DB2, RDS, ANSI SQL.
- Tools and Technologies: Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, Terraform, Linux.
- Concepts: ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/ CD, Microservices, Large-scale data platforms, Infrastructure automation, Privacy-preserving AI, GenAI, Integration Patterns, Data Models, Clinical Care, Networking, Cloud Identity Federation, Orchestration, OIDC.